

Cellular Senescence and FLT1 Signaling in Kidney Cancer: A Multi-Omics and Machine Learning Investigation

<!-- HTML Part 1 (Title -> End of Key Findings) --> <header> <h1>Cellular Senescence and FLT1 Signaling in Kidney Cancer: A Multi-Omics and Machine Learning Investigation</h1> <header> <section aria-labelledby="abstract"> <p> Clear cell renal cell carcinoma (ccRCC) remains one of the most common and aggressive forms of kidney cancer, yet reliable diagnostic markers and targeted treatments are still lacking. A recent study published in The FASEB Journal used an integrated multi-omics and machine learning strategy to explore the role of cellular senescence in ccRCC progression. By combining transcriptomic, proteomic, spatial transcriptomic, and single-cell RNA sequencing data, the researchers identified the FLT1 gene as a central driver of tumor malignancy through its regulation of epithelial-endothelial cell crosstalk. </p> </div> <section> <section aria-labelledby="overview"> <h2 id="overview">Integrating Multi-Omics to Decode Tumor Biology</h2> <p> The research team analyzed large-scale datasets from TCGA, CPTAC, and GEO repositories, encompassing bulk RNA sequencing from over 600 samples, protein-level expression data, and spatial transcriptomic profiles from 24 ccRCC tumors. Single-cell RNA sequencing data from both human and mouse kidney tissues were processed using the Seurat pipeline with Harmony-based batch correction, allowing the identification of five major cell types: endothelial cells, epithelial cells, fibroblasts, immune cells, and others. The AUCel algorithm was applied to score cellular senescence activity across these populations, revealing that senescence levels were broadly elevated in ccRCC tissue compared to normal kidney tissue. A meta-analysis across multiple clinical cohorts confirmed that higher senescence scores were significantly associated with poorer patient survival. </p> </div> <section> <section aria-labelledby="key-findings"> <h2 id="key-findings">Key Findings</h2> Elevated cellular senescence correlates with immune suppression: ccRCC tissues showed significantly higher senescence scores than normal tissue, and increased senescence was linked to a suppressed immune microenvironment and worse patient prognosis across multiple cohorts. FLT1 identified as a pivotal gene through machine learning: Using 101 algorithm combinations and seven independent validation classifiers, the study pinpointed FLT1 as the single most important gene associated with ccRCC progression, with expression roughly seven times higher in tumor tissue than in normal tissue. FLT1-VEGFA-AKT1 axis mediates epithelial-endothelial crosstalk: A regulatory network centered on FLT1 in endothelial cells, VEGFA in epithelial cells, and AKT1 as a downstream effector was shown to drive malignant behavior, with VEGFA-positive proximal tubular epithelial cells representing the most aggressive tumor subtype. Therapeutic and immunotherapy implications: Patients with low FLT1 expression demonstrated better predicted responses to immune checkpoint inhibitors, while molecular docking identified Sorafenib, Regorafenib, and Lenvatinib as candidate FLT1-targeting agents for high-expression patients. </div> <!-- End of Key Findings --> <!-- HTML Part 2 (Figure Caption -> End) --> <section aria-labelledby="figure1"> <figure> <figcaption> Figure 1: Single-cell characterization of cellular senescence in ccRCC. Panels A through C show tSNE representations of major cell types identified by scRNA-seq across all samples, normal tissue, and ccRCC tissue. Panel D displays marker gene expression across five cell types. Panel E illustrates the relative proportions of each cell type. Panel F depicts cellular senescence levels scored using the AUCel algorithm. Panel G shows expression patterns of senescence-related genes. Panels H and I compare senescence scores between normal and ccRCC groups, while Panel J presents a meta-analysis of senescence score and prognosis across multiple cohorts. Adapted from Sun et al. (2026), The FASEB Journal. </figure> <section> <section aria-labelledby="section-one"> <h2 id="section-one">Machine Learning Identifies FLT1 as a Core Regulator</h2> <p> To narrow the field of senescence-associated genes, the authors employed a two-stage machine learning pipeline. In the first stage, 101 combinations of survival modeling algorithms, including LASSO, Ridge regression, random survival forests, and gradient boosting machines, were used to construct prognostic signatures. In the second stage, seven classification algorithms validated FLT1 as the top-performing single gene across training and test partitions using tenfold cross-validation. Proteomic data from the CPTAC database independently confirmed elevated FLT1 protein levels in ccRCC, and experimental knockdown of FLT1 in 786-O cells significantly reduced migration, proliferation, and viability. </p> </div> <section aria-labelledby="section-two"> <h2 id="section-two">Epithelial-Endothelial Crosstalk Through the FLT1 Network</h2> <p> Spatial transcriptomic and single-cell analyses revealed that FLT1 operates within a tightly coordinated signaling axis. Endothelial cells expressing FLT1 and AKT1 interacted with VEGFA-producing proximal tubular epithelial cells, and co-culture experiments confirmed that VEGFA stimulation upregulated FLT1, AKT1, EGFR1, and MMP9 in endothelial populations. Copy number variation analysis verified that FLT1-positive and FLT1/AKT1 double-positive endothelial cells were genetically malignant. This epithelial-endothelial crosstalk mechanism was conserved in mouse ccRCC models, supporting its fundamental biological relevance. </p> </div> <section aria-labelledby="section-three"> <h2 id="section-three">From Biomarker to Therapeutic Target</h2> <p> The translational significance of FLT1 extended to immunotherapy prediction and drug development. Patients with low FLT1 network expression showed higher immune phenotype scores and better predicted responses to anti-PD1 and anti-CTLA4 therapies. For patients with high FLT1 expression, molecular docking and dynamics simulations identified a combination strategy using FLT1 inhibitors such as Sorafenib, Regorafenib, and Lenvatinib, supplemented by the AKT1 inhibitor Capivasertib and the VEGFA inhibitor Bevacizumab. </p> </div> <section aria-labelledby="conclusion"> <h2 id="conclusion">Conclusion</h2> <p> This study presents a comprehensive multi-omics framework that connects cellular senescence to ccRCC malignancy through the FLT1-VEGFA-AKT1 signaling axis. By combining transcriptomics, proteomics, spatial transcriptomics, and single-cell analyses with machine learning, the researchers identified FLT1 as both a prognostic biomarker and a druggable target. These findings offer a foundation for stratifying kidney cancer patients and designing combination therapies tailored to molecular profiles. </p> </div> <section aria-labelledby="citation"> <h2 id="citation">Citation</h2> <p> Sun, M., Mao, H., Luo, Y., Yang, M., He, Z., Li, S., Liu, Z., Peng, L., Zhang, Q., Zhang, J., & Zhang, Y. (2026). Multi-omics and machine learning uncovered FLT1-mediated epithelial-endothelial crosstalk in cellular senescence driving clear cell renal cell carcinoma malignancy. The FASEB Journal, 40(8), e71784. https://doi.org/10.1096/f.202503291RR </p> </div> <section aria-labelledby="note"> <h2 id="note">Note</h2> <p> This blog post summarizes findings from the above-cited research. Figures are adapted from the original publication. For full details, please refer to the source article. </p> </div> <p> By Xinye Yu, Dalton Bioanalytics, specializing in multiomics analysis </p> </div>

URL slug: flt1-cellular-senescence-kidney-cancer-multi-omics

SEO page title: FLT1 and Cellular Senescence in Kidney Cancer: Multi-Omics Review | Scientific Summary

Meta description: A scientific summary of how multi-omics and machine learning uncovered FLT1-mediated epithelial-endothelial crosstalk in cellular senescence driving clear cell renal cell carcinoma malignancy.

Image alt text: Single-cell RNA sequencing tSNE plots and bar charts showing cell type distributions, cellular senescence scoring, and prognostic meta-analysis across normal kidney tissue and clear cell renal cell carcinoma samples.

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